

Validating AI for Image Analysis

Part 2: Further Metrics & Performance Uncertainty

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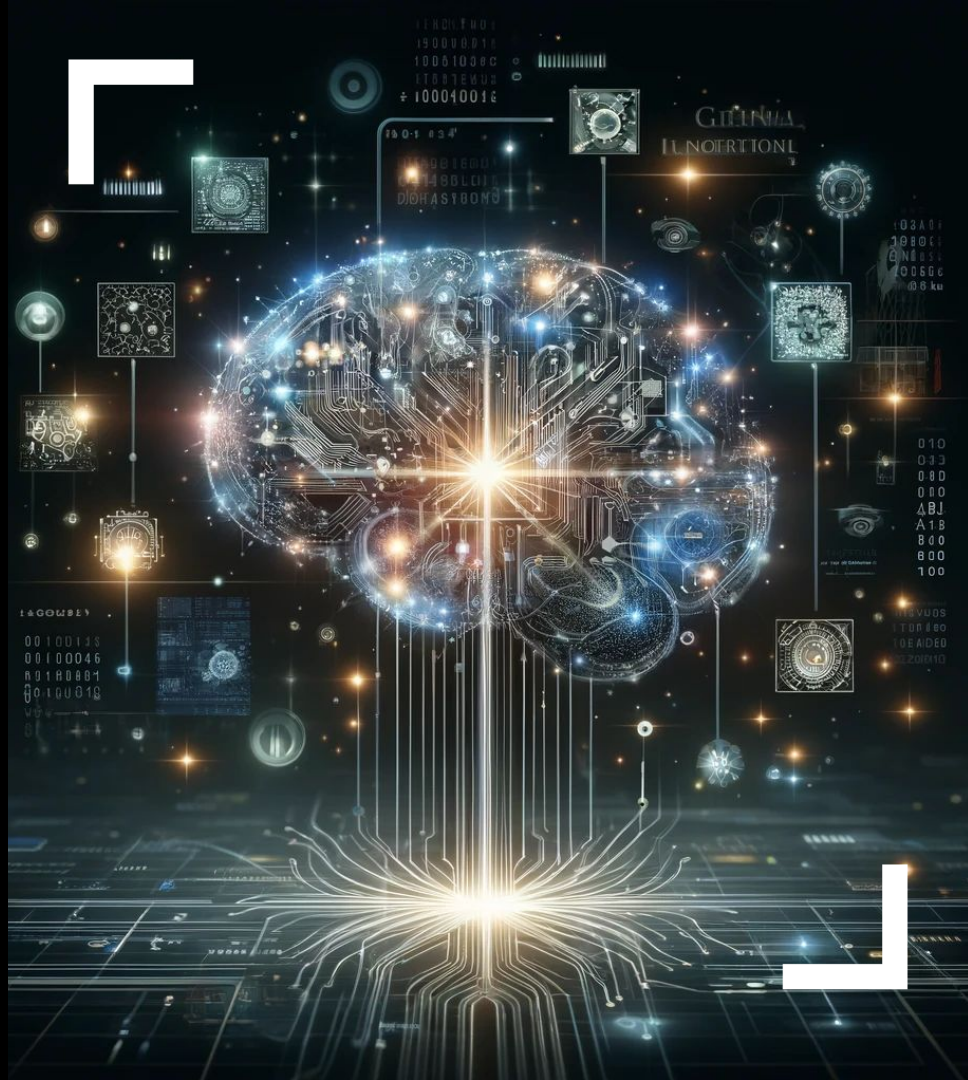
Div. Intelligent Medical Systems division, German Cancer Research Center (DKFZ)

Helmholtz Imaging

 HELMHOLTZ
IMAGING


.....
Research for a Life without Cancer

GERMAN
CANCER RESEARCH CENTER
IN THE HELMHOLTZ ASSOCIATION



Agenda & General Remarks (June 9th 2026)



Time	Topic
09:00	Short Recap of Day 1 + Use Case 4 Pres. + Further Metrics
09:30	Performance Uncertainty Reporting (Part 1)
09:45	1st Practical
10:00	Performance Uncertainty Reporting (Part 2) + 2nd Practical
10:30	Break
10:50	3rd Practical (Breakout Rooms)

Remark 1: If not done so already, please pull from our Github repository

Remark 2: Please keep menti.com open. We will use it throughout the lecture



What happened so far

Metrics we learned about



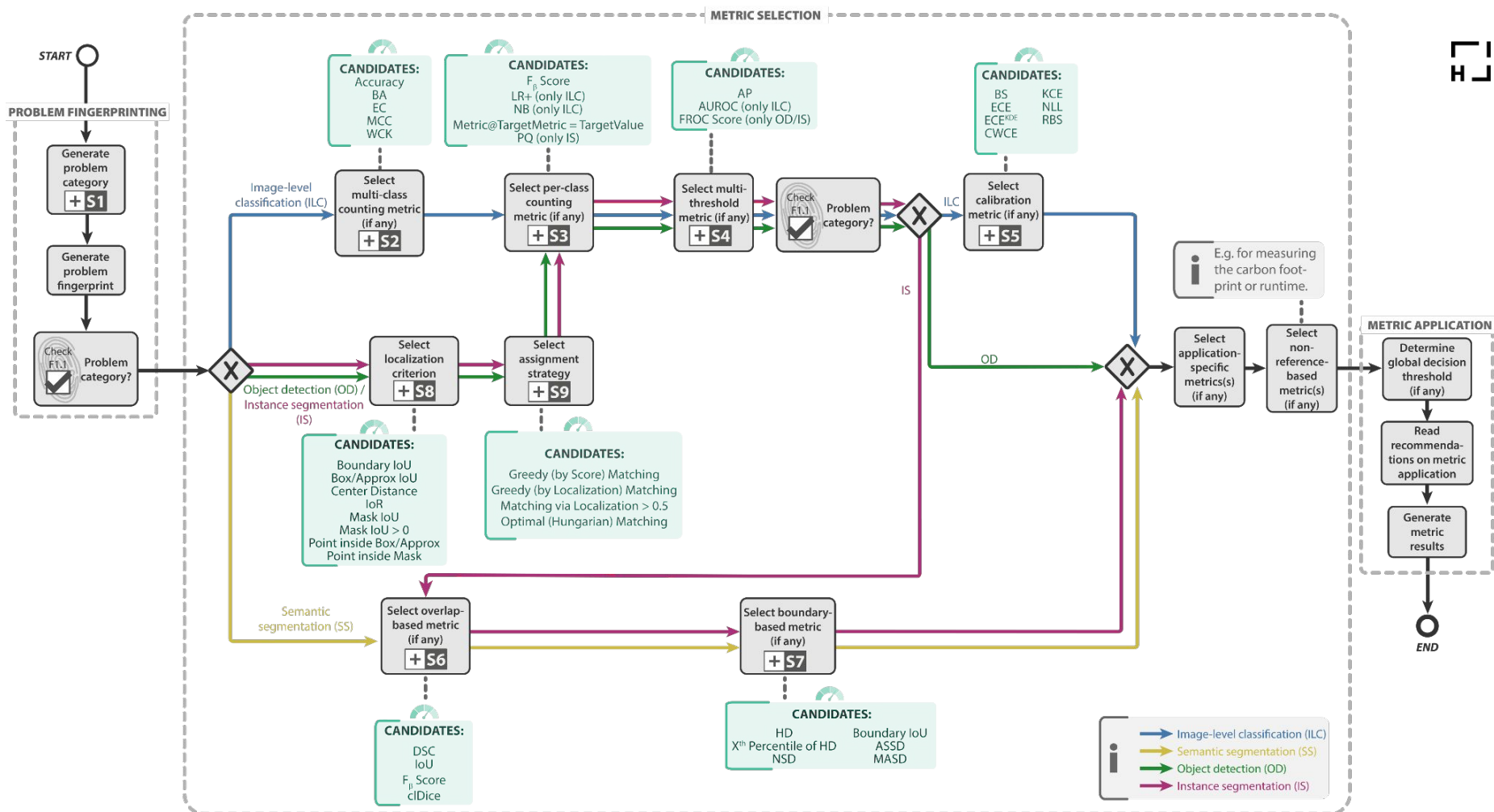
$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN}$$

$$\text{DSC}(A, B) = \frac{2|A \cap B|}{|A| + |B|}$$

$$\text{PPV} = \frac{TP}{TP + FP}$$

$$\text{NSD}(A, B)^{(\tau)} = \frac{\begin{array}{c} \text{blue L-shape} + \text{orange L-shape} \\ \text{blue square} + \text{orange square} \end{array}}{=} = \frac{|S_A \cap \mathcal{E}_B^{(\tau)}| + |S_B \cap \mathcal{E}_A^{(\tau)}|}{|S_A| + |S_B|}$$

$$\text{BA} = \frac{1}{2} \left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP} \right) = \frac{1}{2} (\text{Sensitivity} + \text{Specificity})$$



Use case 4 presentation

Classification metrics

Classification metrics we already talked about



$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN}$$

$$\text{BA} = \frac{1}{2} \left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP} \right) = \frac{1}{2} (\text{Sensitivity} + \text{Specificity})$$

$$\text{PPV} = \frac{TP}{TP + FP}$$

Practical 1



Assumption: We have a method with specific inherent properties:
Sensitivity = 0.90 and Specificity = 0.80

**Performance
on data set 1**

		PREDICTED	
		Positive	Negative
ACTUAL	Positive	TP 45	FN 5
	Negative	FP 10	TN 40

**Performance
on data set 2**

		PREDICTED	
		Positive	Negative
ACTUAL	Positive	TP 81	FN 9
	Negative	FP 2	TN 8

Practical 1



Assumption: We have a method with specific inherent properties:
Sensitivity = 0.90 and Specificity = 0.80

**Performance
on data set 1**

		PREDICTED	
		Positive	Negative
ACTUAL	Positive	TP 45	FN 5
	Negative	FP 10	TN 40

Accuracy = 0.85
BA = 0.85
PPV = 0.82
EC = 0.15

**Performance
on data set 2**

		PREDICTED	
		Positive	Negative
ACTUAL	Positive	TP 81	FN 9
	Negative	FP 2	TN 8

Accuracy = 0.89
BA = 0.85
PPV = 0.98
EC = 0.15

Prevalence



The prevalence is the proportion of actual positive samples in a data set

**Confusion
matrix 1**

		PREDICTED	
		Positive	Negative
ACTUAL	Positive	TP 45	FN 5
	Negative	FP 10	TN 40

**Confusion
matrix 2**

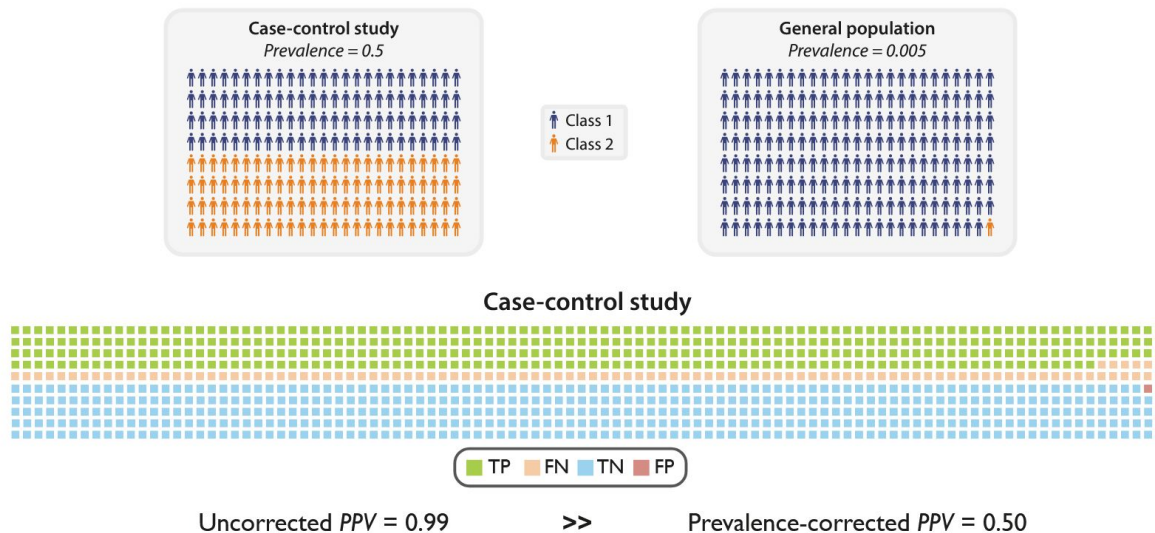
		PREDICTED	
		Positive	Negative
ACTUAL	Positive	TP 81	FN 9
	Negative	FP 2	TN 8

$$\frac{TP + FN}{TP + FN + TN + FP}$$

Prevalence-dependent metrics



Some metrics depend on the prevalence. They should not be used to compare datasets with varying prevalences.



$$\text{PPV}_{\text{corrected}} = \frac{\text{Sensitivity} \cdot \text{Prevalence}}{\text{Sensitivity} \cdot \text{Prevalence} + (1 - \text{Specificity}) \cdot (1 - \text{Prevalence})}$$

Expected Cost (EC)



- Generalization of the probability of error ($1 - \text{Accuracy}$) for cases with different error consequences
- Defined as the expectation of the cost
 - In practice, the expectation can be estimated as an average of the costs over the evaluation samples
- EC describes the weighted sum of error rates
- It depends on the prevalence and can therefore be prevalence-corrected

Expected Cost (EC)

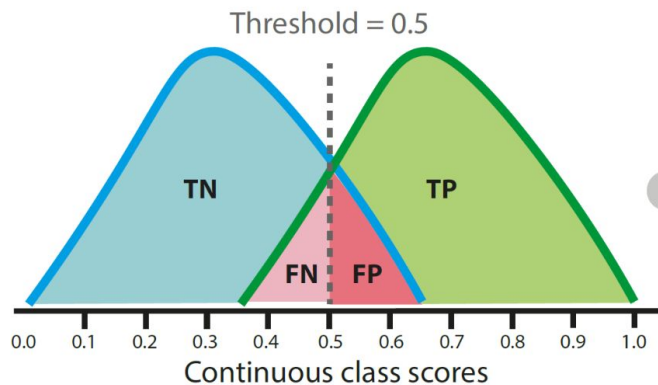


$$EC = w_{\text{miss}} \cdot \underbrace{\frac{FN}{TP + FN}}_{\text{Miss rate}} \cdot \underbrace{\frac{TP + FN}{TP + TN + FP + FN}}_{\text{Prevalence}} + w_{\text{FA}} \cdot \underbrace{\frac{FP}{TN + FP}}_{\text{False alarm rate}} \cdot \left(1 - \underbrace{\frac{TP + FN}{TP + TN + FP + FN}}_{\text{Prevalence}}\right)$$

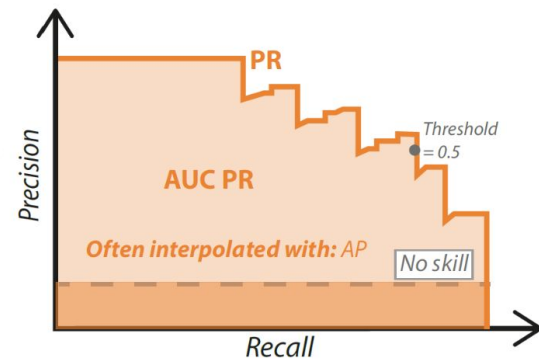
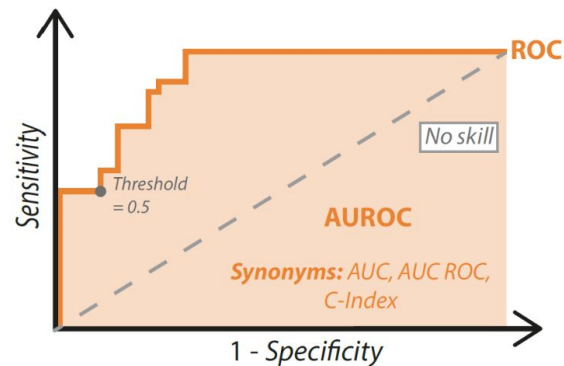
$w_{\text{miss}} / w_{\text{FA}}$: Estimation of costs of the respective errors

- Depending on how the costs are chosen, EC can be seen as a generalization of several other metrics; e.g.:
 - Accuracy = 1 - EC for 0-1 costs

Multi-threshold metrics



Scan over
thresholds

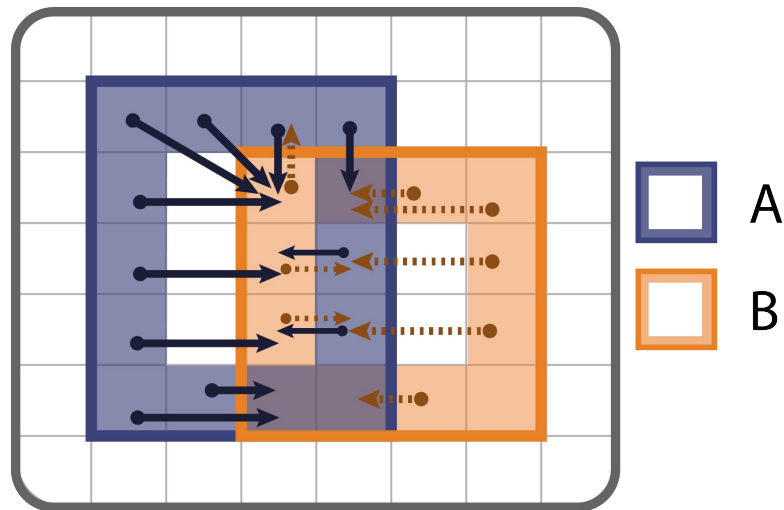


Segmentation metrics

Average Symmetric Surface Distance (ASSD)



Boundary-based segmentation metric



→ Min. distances from boundary pixels in A to B

→ Min. distances from boundary pixels in B to A

"ASSD measures the average of all shortest boundary distances between contour A to any point on contour B and vice versa, symmetrically"

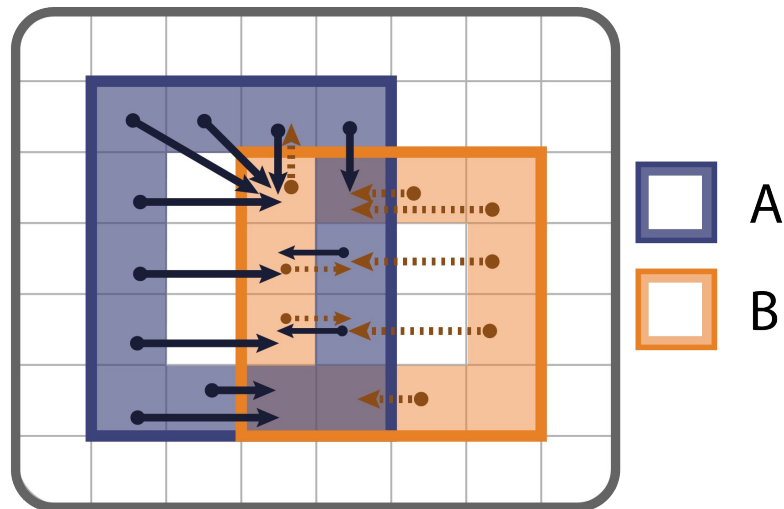
$$\text{ASSD}(A,B) = \left\{ \begin{array}{c} \text{solid arrows} \\ \text{dotted arrows} \end{array} \right\}$$

average

Average Symmetric Surface Distance (ASSD)



Boundary-based segmentation metric



→ Min. distances from boundary pixels in A to B

→ Min. distances from boundary pixels in B to A

"ASSD measures the average of all shortest boundary distances between contour A to any point on contour B and vice versa, symmetrically"

$$\text{ASSD}(A,B) = \left\{ \begin{array}{c} \text{solid arrows} \\ \text{dotted arrows} \\ \text{solid arrows} \end{array} \right\}$$

average

Performance Uncertainty

Performance variability reporting is crucial for clinical translation



The **statistical analysis plays a critical role** in the assessment of [...] ML performance but may be under-appreciated by many ML developers. [...] There are still publications that present point estimates of ML performance without quantification of uncertainties.

A Regulatory Science Perspective on Performance Assessment of Machine Learning Algorithms in Imaging



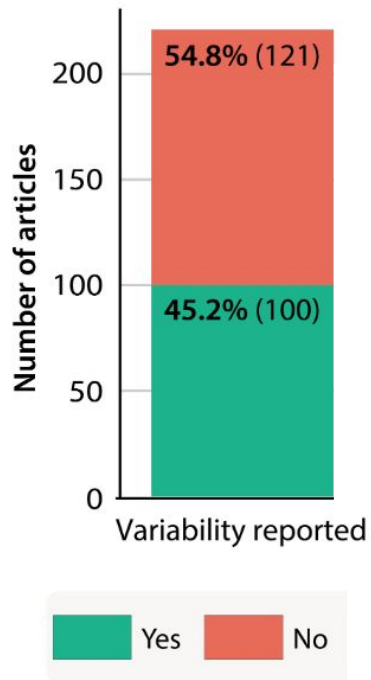
Commonly encountered results tables

Methods		DSC	HD95
Method 1		93.82	43.22
Method 2		92.64	63.68
Method 3		94.67	67.34
Proposed method		94.70	67.35

Do we have a problem in practice?



(analysis of all >200 MICCAI 2023 segmentation papers)



Computing the mean value: on which dataset?



Methods	DSC	HD95
Method 1	0.892	1.23
Method 2	0.895	1.22
Method 3	0.883	1.32
Proposed	0.897	1.21

Data splitting



Whole dataset



Validation set

Single split

Data splitting



Whole dataset



Validation set



Validation set



Validation set



Validation set

Cross-validation

Data splitting



Whole dataset



Validation set



Validation set



Validation set



Validation set

✓ Use to tune hyperparameters, experiment with different architectures...

✗ Do not use to report final performance (biased)

Data splitting



Whole dataset



Validation set

Test set



Validation set



Validation set



Validation set

✓ Use to report final performance

✓ Use to tune hyperparameters, experiment with different architectures...

✗ Do not use to report final performance (biased)

Data splitting



Whole dataset



Validation set

Test set



Validation set



Validation set



Generalization set 1
(from another dataset)



Validation set



Generalization set 2
(from yet another dataset)

✓ Use to tune hyperparameters, experiment with different architectures...

✗ Do not use to report final performance (biased)

Data splitting



Whole dataset



Validation set

Test set



Validation set



Validation set



Validation set

✓ Use to tune hyperparameters, experiment with different architectures...

✗ Do not use to report final performance (biased)



Gen
(from

Study OOD
performance

Gen
(from yet another dataset)

Computing the mean value: on which dataset?



Mean DSC and HD95 *on the test set*

Methods	DSC	HD95
Method 1	0.892	1.23
Method 2	0.895	1.22
Method 3	0.883	1.32
Proposed	0.897	1.21

Paper includes text describing precisely the data splitting and which splits were used for what purpose

Summary Statistics

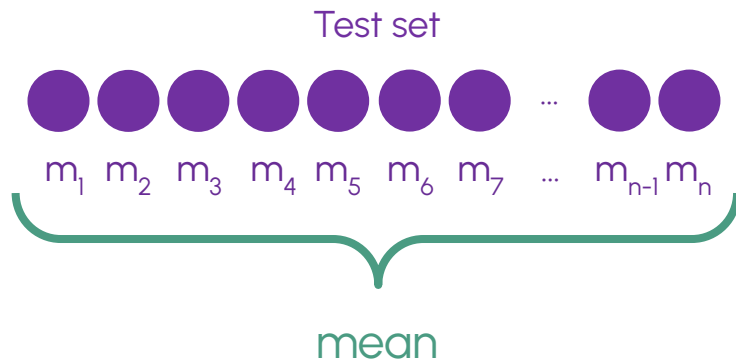


Mean DSC and HD95 on the test set

Methods	DSC	HD95
Method 1	0.892	1.23
Method 2	0.895	1.22
Method 3	0.883	1.32
Proposed	0.897	1.21

Individual
values of
the metric

e.g. DSC
on each
individual



Summary Statistics

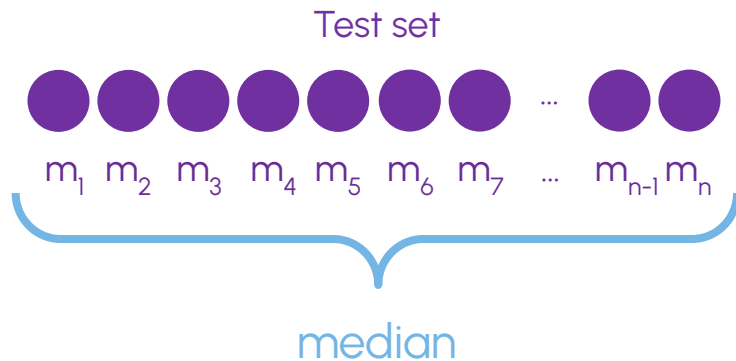


Median DSC and HD95 on the test set

Methods	DSC	HD95
Method 1	0.892	1.23
Method 2	0.895	1.22
Method 3	0.883	1.32
Proposed	0.897	1.21

Individual
values of
the metric

e.g. DSC
on each
individual

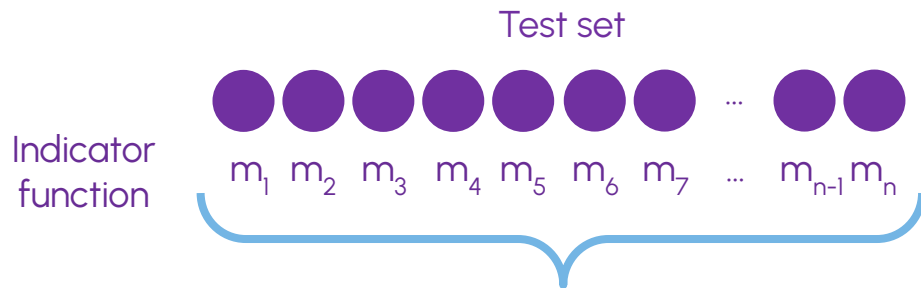


! Some metrics are only defined on a set



Accuracy and AUC **on the test set**

Methods	Accuracy	AUC
Method 1	0.828	0.862
Method 2	0.821	0.857
Method 3	0.847	0.889
Proposed	0.851	0.891



! Important implications for variability

What do we mean by SD of accuracy?
SD of its **sampling distribution**

Reporting variability: which variability?



$\pm$ what?

At least 3 possibilities

Methods	DSC
Method 1	0.892 $\pm$ 0.017
Method 2	0.895 $\pm$ 0.013
Method 3	0.883 $\pm$ 0.012
Proposed	0.897 $\pm$ 0.013

Standard-deviation (SD) of
the metric over the test set

2 Standard-error (SE) of the
summary statistic

3 Standard-deviation (SD) over
cross-validation (CV)

Reporting variability: which variability?



$\pm$ what?

At least 3 possibilities

Methods	DSC
Method 1	0.892 $\pm$ 0.017
Method 2	0.895 $\pm$ 0.013
Method 3	0.883 $\pm$ 0.012
Proposed	0.897 $\pm$ 0.013

Standard-deviation (SD) of
the metric over the test set

2 Standard-error (SE) of the
summary statistic

3 Standard-deviation (SD) over
cross-validation (CV)

Standard-deviation (SD)

SD of your metric across individuals
(e.g. over test set)

 **Meaning: How variable is your performance across your set**

Its magnitude is independent of n

 Descriptive statistic

Standard error (SE)

SD of the sampling distribution of
a statistic (e.g. over test set)

 **Meaning: How precise the estimate of the statistic**

Shrinks with $\sqrt{n}$

 Inferential statistic

Practical 2.1

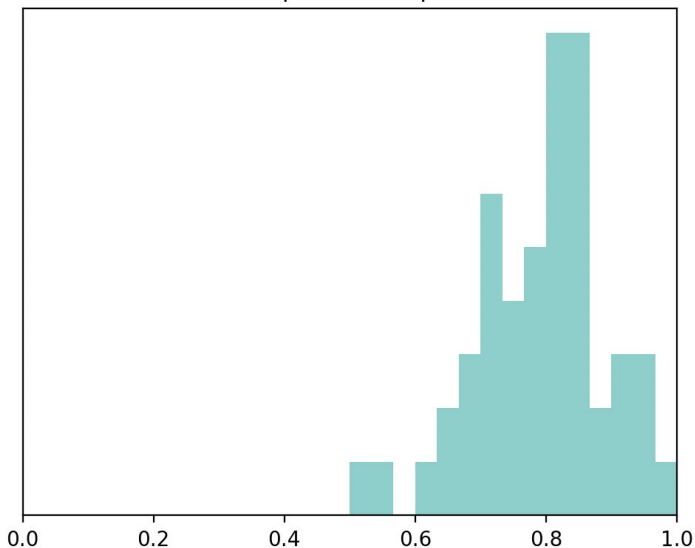
- Standard Deviation (SD)
 - $SD = \sqrt{[\sum_{i=1}^n (x_i - \bar{x})^2 / (n-1)]}$
- Standard Error (SE)
 - $SE = SD / \sqrt{n}$
- Parameters
 - x_i = individual observation
 - $\bar{x}$ = sample mean
 - n = sample size
 - SD = sample standard deviation
 - Σ = sum over all observations

Sampling Distribution



Distribution of a statistic across random samples

Empirical Sample



OK but I have only
one dataset!

Parametric methods

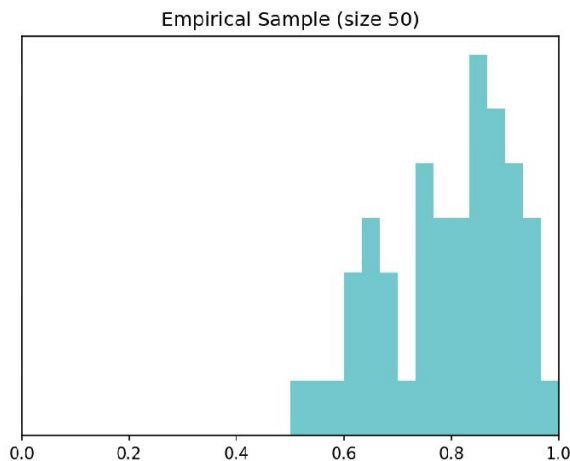
Underlying distribution is
known (e.g. Gaussian)

Or asymptotic results

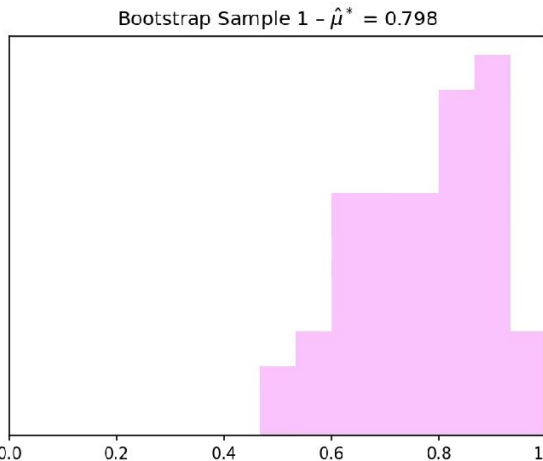
Non-parametric
methods

In particular the bootstrap

Bootstrap: Approximating the sampling distribution

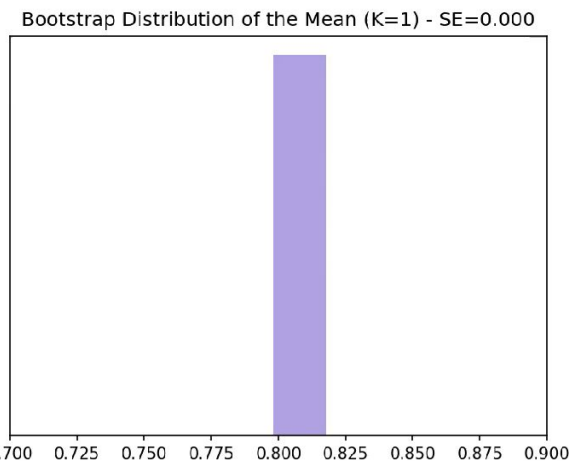


You have a sample of size n



Generate bootstrap samples

- Randomly draw n values **with replacement** from your sample
- Repeat this process many times (e.g., 9999 times)
- Each time, compute the statistic of interest (e.g., the mean) on the bootstrap sample



These values form the bootstrap distribution

This is an **approximation of the sampling distribution** of your statistic.

Back to FDA recommendations: Confidence Intervals



Methods	DSC	HD95
Method 1	79.9 [76.6, 82.2]	8.05 [6.85, 9.37]
Method 2	79.7 [76.4, 82.3]	8.11 [6.93, 9.42]
Method 3	80.1 [76.9, 82.5]	7.91 [6.71, 9.22]
Proposed	80.2 [77.1, 82.6]	7.73 [6.65, 8.91]

[...] All performance estimates should be provided with confidence intervals [...]

FDA-2024-D-4488: Artificial Intelligence-Enabled Device
Software Functions: Lifecycle Management and Marketing
Submission Recommendations



U.S. FOOD & DRUG
ADMINISTRATION

Confidence Intervals



→ Need to be computed from independent test set

Various methods including

1

Parametric methods

✓ Theoretical guarantees **when
distributional assumptions met**

✗ Each summary statistic
requires special treatment

2

Bootstrap

✗ Less theoretical guarantees

✓ No distributional
assumptions

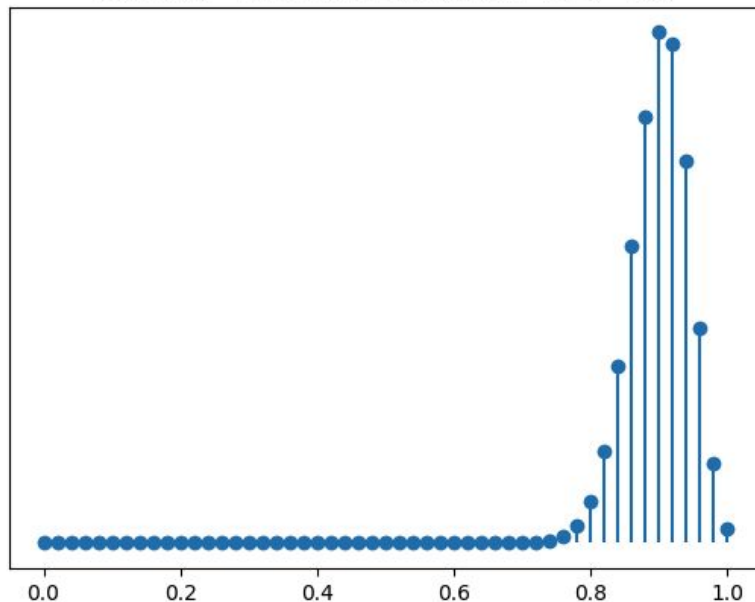
✓ Can be applied to many
summary statistics

Confidence Intervals



Metric distribution
✓ In **a few** cases, it is known

Accuracy - Binomial Proportion ($n=50$, $p=0.9$)

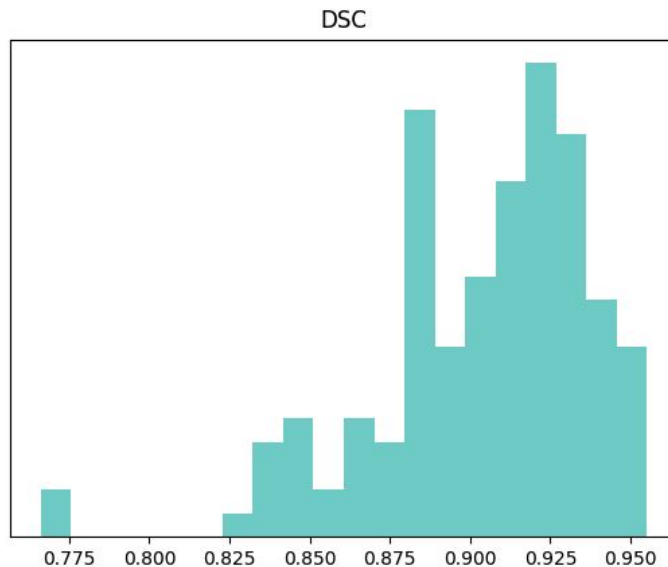


Accuracy follows a
binomial
proportion

Confidence Intervals



Metric distribution
✗ In **most** cases, it is not

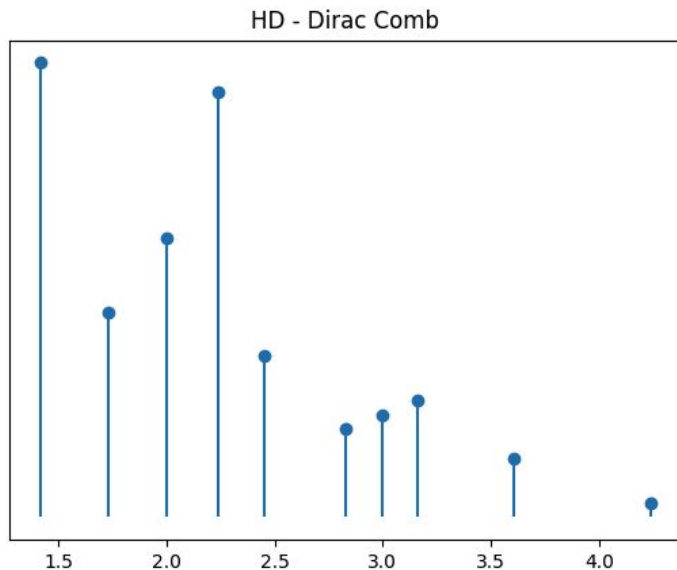


Some are
semi-continuous

Confidence Intervals



Metric distribution
✗ In **most** cases, it is not



Some are discrete

Practical 2.2

Results (1) - no variability (classification)



Test set, n = 20

Algorithm	Accuracy
alg_01	0.800
alg_02	0.800
alg_03 (proposed)	0.850
alg_04	0.750
alg_05	0.800

Test set, n = 1000

Algorithm	Accuracy
alg_01	0.800
alg_02	0.802
alg_03 (proposed)	0.822
alg_04	0.798
alg_05	0.805

Results (2) - no variability (segmentation)



Test set, n = 50

Algorithm	DSC
alg_01	0.790
alg_02	0.795
alg_03 (proposed)	0.807
alg_04	0.792
alg_05	0.798

Test set, n = 5000

Algorithm	DSC
alg_01	0.798
alg_02	0.801
alg_03 (proposed)	0.822
alg_04	0.800
alg_05	0.803

Results (3) - with CIs (classification)



Test set, n = 20

Algorithm	Accuracy	Mean Acc across bootstrap resamplings (95% CI)
alg_01	0.800	0.800 (0.600, 0.950)
alg_02	0.800	0.796 (0.600, 0.950)
alg_03 (proposed)	0.850	0.850 (0.650, 1.000)
alg_04	0.750	0.753 (0.550, 0.950)
alg_05	0.800	0.796 (0.600, 0.950)

Test set, n = 1000

Algorithm	Accuracy	Mean Acc across bootstrap resamplings (95% CI)
alg_01	0.800	0.800 (0.776, 0.825)
alg_02	0.802	0.802 (0.776, 0.824)
alg_03 (proposed)	0.822	0.822 (0.798, 0.845)
alg_04	0.798	0.798 (0.773, 0.824)
alg_05	0.805	0.804 (0.781, 0.828)

Results (4) - with CIs (segmentation)



Test set, n = 50

Algorithm	DSC	Mean DSC across bootstrap resamplings (95% CI)
alg_01	0.790	0.790 (0.777, 0.803)
alg_02	0.795	0.795 (0.782, 0.808)
alg_03 (proposed)	0.807	0.807 (0.793, 0.821)
alg_04	0.792	0.792 (0.779, 0.805)
alg_05	0.798	0.798 (0.785, 0.811)

Test set, n = 5000

Algorithm	DSC	Mean DSC across bootstrap resamplings (95% CI)
alg_01	0.800	0.800 (0.776, 0.825)
alg_02	0.802	0.802 (0.776, 0.824)
alg_03 (proposed)	0.822	0.822 (0.798, 0.845)
alg_04	0.798	0.798 (0.773, 0.824)
alg_05	0.805	0.804 (0.781, 0.828)

Practical 2.3

CIs of the difference - classification



Test set, n = 20

Comparison	Mean difference across bootstrap resamplings (95% CI)
alg_03 - alg_01	0.050 (-0.150, 0.250)
alg_03 - alg_02	0.054 (-0.200, 0.300)
alg_03 - alg_04	0.098 (-0.150, 0.300)
alg_03 - alg_05	0.054 (-0.200, 0.300)

Test set, n = 1000

Comparison	Mean difference across bootstrap resamplings (95% CI)
alg_03 - alg_01	0.022 (-0.011, 0.056)
alg_03 - alg_02	0.021 (-0.012, 0.055)
alg_03 - alg_04	0.024 (-0.012, 0.058)
alg_03 - alg_05	0.018 (-0.014, 0.053)

CIs of the difference - segmentation

Test set, n = 50

Comparison	Mean difference across bootstrap resamplings (95% CI)
alg_03 - alg_01	0.017 (0.011, 0.023)
alg_03 - alg_02	0.012 (0.005, 0.019)
alg_03 - alg_04	0.015 (0.008, 0.022)
alg_03 - alg_05	0.009 (0.003, 0.016)

Test set, n = 5000

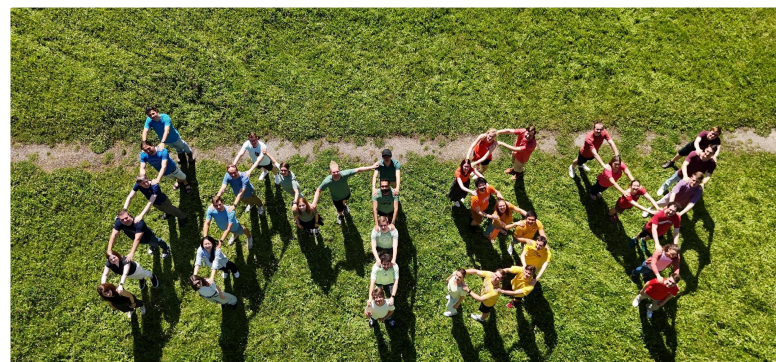
Comparison	Mean difference across bootstrap resamplings (95% CI)
alg_03 - alg_01	0.024 (0.023, 0.025)
alg_03 - alg_02	0.021 (0.020, 0.022)
alg_03 - alg_04	0.022 (0.021, 0.023)
alg_03 - alg_05	0.019 (0.018, 0.020)

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